SMART MULTI-SENSOR WEARABLE FOR REAL-TIME ELDERLY HEALTH ANOMALY AND FALL DETECTION

COMMON INTEGRATED SOLUTION REPORT

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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ABSTRACT

Older adults who live at home often deteriorate without anyone noticing: a heart rhythm disturbance passes in minutes, oxygen saturation drifts down overnight, stress builds up unspoken, or a fall leaves the person unable to call for help. Clinic visits capture only a moment, and most consumer wearables track single signals or decide in the cloud. This report presents SmartGlove, the integrated solution developed by Group R26-IT-108 for the project "Smart Multi-Sensor Wearable for Real-Time Elderly Health Anomaly and Fall Detection". The system combines a sensor glove, a waist-worn belt and a local hub computer, so detection never depends on the internet. Four components address four parts of the problem: beat-level arrhythmia screening from a single-lead ECG, on-device fall detection and posture classification in the belt, physiological and facial stress screening, and hypoxemia, sepsis and Parkinson's tremor screening with an SOS gesture. The belt's posture is shared as context for heart-rate alerts, and the outputs of all screening engines are merged into one Master Risk Score from 0 to 100, with emergency alerts and the user's location sent through Firebase to a carer's mobile application. Each component was evaluated with held-out data or physical trials: the belt detected 46 of 48 simulated falls with 2 false alarms in 72 daily activities, fall alerts reached a phone in 23 of 24 end-to-end tests with a mean delay of 1.8 s, and the vital-sign screening models ran in under 4 ms each on a CPU, with accuracies from 80.7% (arrhythmia) to 98.9% (hypoxemia). Most models were trained on public data whose origin is synthetic or undocumented, so these figures show technical feasibility rather than clinical accuracy. The project shows that a low-cost, explainable, locally processed multi-sensor wearable is practical, and identifies validation with older adults as the essential next step.

Keywords: Elderly health monitoring, Wearable IoT, Fall detection, Edge computing, Sensor fusion

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AAMI	Association for the Advancement of Medical Instrumentation
AF	Atrial Fibrillation
BPM	Beats Per Minute
CPU	Central Processing Unit
ECG	Electrocardiogram
EQ	Evaluation Question
GO	Group Objective
GPS	Global Positioning System
GSR	Galvanic Skin Response
HR	Heart Rate
HTTP	Hypertext Transfer Protocol
IEC	International Electrotechnical Commission
IMU	Inertial Measurement Unit
IoT	Internet of Things
IP	Intellectual Property
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
LGPL	GNU Lesser General Public License
NMRA	National Medicines Regulatory Authority
OLED	Organic Light-Emitting Diode
PDPA	Personal Data Protection Act
SDG	Sustainable Development Goal
SIRS	Systemic Inflammatory Response Syndrome
SLIIT	Sri Lanka Institute of Information Technology
SMV	Signal Magnitude Vector
SOS	Emergency distress signal
SP	Subproblem
SpO2	Peripheral Oxygen Saturation
SSE	Server-Sent Events
SVEB	Supraventricular Ectopic Beat
UDP	User Datagram Protocol
USD	United States Dollar
VEB	Ventricular Ectopic Beat
WHO	World Health Organization

1 INTRODUCTION AND PROJECT CONTEXT

This chapter introduces the domain of the project, the people it is meant to serve, its relationship to the Sustainable Development Goals, and the purpose of this common report in relation to the four individual theses.

1.1 Background and context

The world's population is ageing quickly. The World Health Organization estimates that by 2030 one person in six will be aged 60 or over, and that the number of people in this age group will reach about 2.1 billion by 2050 [1]. Sri Lanka is ageing faster than most countries in South Asia, and the World Bank has noted that the share of older people in the population is approaching that of high-income countries while family support structures are changing [2]. Many older Sri Lankans live with adult children who work during the day, or live alone, so there are long periods when nobody is watching.

Several common health events in older adults share one dangerous property: they can happen without anyone noticing. Cardiovascular disease is the leading cause of death worldwide, responsible for about 17.9 million deaths in 2019 [3], and many rhythm disturbances are brief and missed by a short clinic ECG [4]. Oxygen saturation can fall to dangerous levels without obvious breathlessness, a pattern described as silent hypoxemia [5]. Sepsis, a life-threatening response to infection, develops over hours and is easily mistaken for ordinary illness at first [6]. Falls are frequent and serious: in people with Parkinson's disease, about half fall within six months [7], and a person who cannot get up may lie on the floor for a long time. Stress and low mood are rarely measured at all in older adults, except through occasional questionnaires.

Wearable sensors make continuous monitoring possible, and the market for wearable health devices has grown rapidly [8], [9]. However, most consumer devices track a few signals, process the data on a phone or in the cloud, and do not share information between sensors. Clinical instruments such as Holter monitors are accurate but analyse the data only after the recording ends [10]. The setting for this project is therefore the

home or care home, where an inexpensive system has to watch several signals at once, decide locally and quickly, and call a carer when something goes wrong.

1.2 Stakeholders and user needs

The stakeholders of the system were identified from the four components' requirements and from discussions within the group and with the supervisors. Table 1.1 summarises them.

Table 1.1: Stakeholders of the integrated solution and their main needs

Stakeholder	Type	Role	Main needs
Older adults, including people with Parkinson's disease	Primary	Wear the glove and belt during daily life	Comfortable, no manual operation, few false alarms, help when unable to call
Family members and home carers	Primary	Receive alerts on the mobile application	Timely alerts that say what happened and where
Care home staff and nurses	Secondary	Monitor several residents	One dashboard, clear priorities, alerts that can be dismissed
Doctors	Secondary	Review alert history	Standard measures, honest accuracy, explainable outputs
Care homes and home-care providers	Secondary	Possible buyers	Low cost per resident, works without reliable internet

Source: Group analysis, based on the four individual theses.

Five user needs follow from this analysis and run through all four components:

- **N1 Continuous, passive monitoring.** The user should not have to start a measurement or press a button for the system to work.
- **N2 Fast alerts that reach a carer.** An emergency should reach a carer's phone within seconds, with the user's location.
- **N3 Few false alarms.** Alerts must be trustworthy, or carers will stop responding to them.
- **N4 Operation without reliable internet.** Detection must not stop when the connection drops.
- **N5 Privacy and explainability.** Raw health data should stay local where possible, and alerts should be explainable in plain terms.

1.3 Alignment with the Sustainable Development Goals

The project relates mainly to Sustainable Development Goal (SDG) 3, Good Health and Well-being, which aims to ensure healthy lives and promote well-being for all at all ages [11]. Within it, target 3.4 seeks to reduce premature mortality from noncommunicable diseases by one third and to promote mental health and well-being; the arrhythmia, respiratory and stress components relate directly to this target. The project also relates to SDG 10, Reduced Inequalities, in the sense that it uses low-cost components and local processing, which could make continuous monitoring available to households and care homes that cannot afford clinical equipment or reliable internet. These are intended contributions; the project has not measured any health or social impact, and Section 7.5 discusses what would be needed to do so.

1.4 Purpose and scope of the common report

This report presents the common problem, the integrated solution and its overall evaluation. It explains how the four components divide the problem between them, how they share data and services, how the complete system behaves, and what the team learned as a whole. It does not repeat the detailed research of each component. The individual theses provide the component-level literature reviews, methods, implementation details and experiments [12], [13], [14], [15]. Where this report summarises a component result, the reader is referred to the relevant thesis for the full evidence.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

This chapter states the common problem, its root causes and the overall research gap, explains how the problem divides into four subproblems handled by the four components, and sets out the group's aim, objectives, scope and constraints.

2.1 Overall problem definition

Older adults living at home or in care homes experience health events, such as heart rhythm disturbances, falls in oxygen saturation, early signs of infection, tremor-related decline, stress and falls, that develop or occur when nobody is watching. Current home monitoring relies on periodic clinical checks, single-purpose devices and manual alarm buttons. As a result, many of these events are noticed late or not at all, and help arrives late, particularly when the person cannot raise the alarm.

Stated precisely, the common problem is: **how can several physiological and physical warning signs in an older adult be watched continuously at home, interpreted together, and turned into timely, trustworthy alerts for carers, at low cost and without depending on internet connectivity for the decision?** This statement describes the need and does not assume any particular technology.

2.2 Root causes and overall research gap

The group's literature reviews identified five root causes.

1. **Episodic measurement.** Clinic visits and spot checks capture a few minutes of data, while arrhythmias, desaturation and falls happen at unpredictable times [4], [5].
2. **Dependence on the user.** Alarm pendants and buttons rely on the person being conscious and able to press them, and many older people who own a call alarm do not use it after a fall [13].

3. **Single-signal devices.** Most wearables monitor one or two signals in isolation. A fast heart rate, for example, cannot be interpreted without knowing whether the person is walking or lying down.
4. **Cloud dependence.** Many systems process data on a phone or a remote server, which adds delay, fails without internet access and moves sensitive health data off the premises [16].
5. **Closed, unexplained decisions.** Commercial devices use proprietary algorithms that carers and clinicians cannot inspect, which makes it harder to trust or tune the alerts.

The overall research gap is the lack of an integrated, low-cost system that covers several health risks of older adults at once, shares context between sensors so that one sensor's information improves the interpretation of another's, makes its decisions locally and explainably, and delivers one coherent alert stream to carers. Each individual thesis identifies a narrower gap within its own area [12], [13], [14], [15]; this report is concerned with the gap that remains when those components stand alone.

2.3 Subproblem and component mapping

The common problem was divided into four subproblems that do not overlap but depend on each other, as shown in Figure 2.1 and Table 2.1. Each subproblem concerns a different kind of warning sign, needs different sensors and methods, and was owned by one member. All four rely on the same hub, posture context, risk score and alert channel, which were built together.

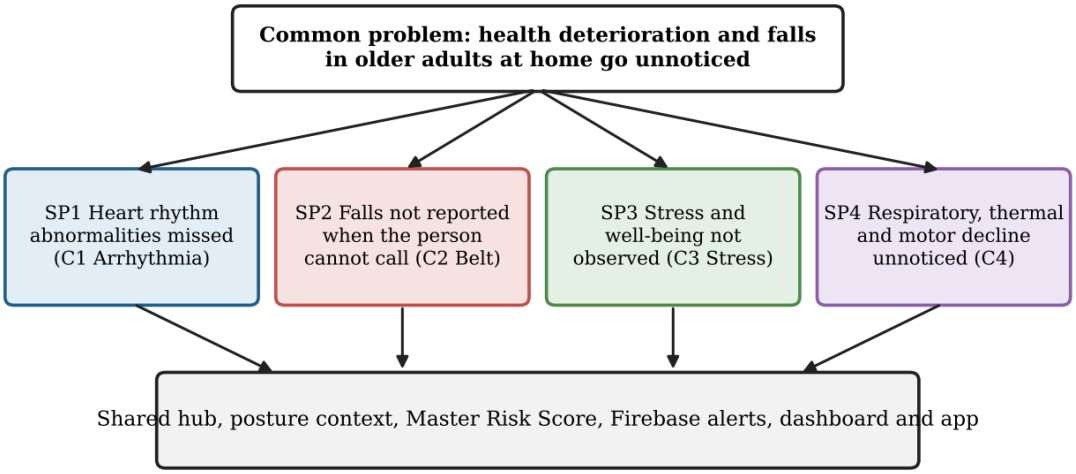


Figure 2.1: The common problem, its four subproblems and the shared integration layer

Source: Group.

Table 2.1: Subproblems, components and owners

Subproblem	Component	Owner	Why it is needed
SP1 Heart rhythm abnormalities go unnoticed	C1 Arrhythmia screening from wearable ECG	Abeysinghe S.A.M.M. (IT22081148)	Brief arrhythmias are missed by spot checks; no other component reads the ECG
SP2 Falls are not reported when the person cannot call	C2 Waist-worn Smart Belt for fall detection	Lakshan N K T (IT22196156)	Falls are the most urgent event; the belt also supplies posture to all other components
SP3 Stress and well-being are not observed	C3 Stress Monitoring System	E.A.P. Peshala (IT22146410)	Adds the psychological dimension that vital-sign screening alone does not cover
SP4 Respiratory, thermal and motor decline goes unnoticed	C4 Respiratory and Thermal Warning System	Dissanayaka D.M.S.K. (IT22291196)	Covers low oxygen, early sepsis and tremor, plus the user-initiated SOS gesture

Source: Group.

The dependencies between components are what make the four a single solution. The belt's posture is used by the hub to decide whether a fast heart rate measured by the glove is explained by activity, which affects the alerts of C1 and C4. C1, C3 and C4 all feed their likelihoods into the same Master Risk Score. C2's fall flag and C4's SOS gesture both force an immediate emergency through the same alert pipeline. Without

the shared layer, each component would be a separate device with its own alarm, which is exactly the fragmentation the common problem describes.

2.4 Overall aim and group objectives

The common aim of the project was to design, implement and evaluate a low-cost, multi-sensor wearable system that continuously monitors older adults for cardiac, respiratory, thermal, motor and stress-related warning signs and for falls, interprets these signals together on a local hub, and delivers timely, explainable alerts to carers without relying on the internet for detection. Table 2.2 lists the group objectives, the components involved and the evidence used to evaluate each one.

Table 2.2: Group objectives, contributing components and evaluation evidence

ID	Group objective	Components	Evaluation evidence
GO1	Acquire ECG, pulse oximetry, temperature, motion and GSR on the glove and motion on the belt, and stream them to a local hub	All	Telemetry received at the hub; component sensor tests
GO2	Screen for arrhythmia, hypoxemia, sepsis, tremor and stress with models that run on a CPU in real time	C1, C3, C4	Held-out model metrics; inference time
GO3	Detect falls on the wearable and share posture with the rest of the system	C2	Fall and activity trials; posture accuracy
GO4	Use posture context to suppress activity-explained heart-rate alerts	C1, C2	Unit and scenario tests of the fusion logic
GO5	Combine all engines into one risk score and deliver emergency alerts with location to carers	All	End-to-end alert tests; delivery time
GO6	Keep detection local, protect health data and keep decisions explainable	All	Design review; feature-based and rule-based outputs

Source: Group.

2.5 Scope assumptions and constraints

Scope. The solution covers the glove and belt hardware and firmware, the hub backend with its screening engines, sensor fusion, correlation engine and Master Risk Score, the Firebase synchronisation, the web dashboard and the carer mobile application.

Clinical diagnosis, treatment decisions, long-term storage of medical records and certification as a medical device are outside the scope.

Assumptions. The wearer puts on both the glove and the belt in the intended position; the glove's ECG electrodes and GSR electrodes make contact; the glove, belt and hub are on the same Wi-Fi network; and the hub, backend and Firebase are running when an alert has to be delivered.

Constraints. The project worked within a student budget and an academic year. No clinical datasets with expert labels or ethical approval for patient trials were available, so the screening models were trained on public datasets and the fall trials used healthy adult volunteers performing simulated falls onto a mattress. The hub had no graphics processing unit, so every model had to run on a CPU within a few milliseconds.

Dependencies. Detection of falls and SOS gestures happens on the wearables and the hub, but alert delivery to carers depends on internet access for Firebase and the phone. All four components depend on the shared telemetry format and on the hub being available.

3 OVERALL SOLUTION STRUCTURE

This chapter presents SmartGlove as one system. It gives an overview of the integrated solution, its high-level architecture, a summary of each of the four components in a common format, and the end-to-end flow of data through the system.

3.1 Integrated solution overview

SmartGlove is a wearable monitoring system made of two worn devices and one local computer. The smart glove measures the user's ECG, heart rate, blood oxygen saturation, skin temperature, hand motion, galvanic skin response and location. The waist-worn Smart Belt detects falls on the device and reports the wearer's posture. Both send their data over Wi-Fi to a hub computer in the home or care home, which runs every screening model and rule, combines their outputs, and decides whether to raise an alert. Alerts, with the user's location, are synchronised through Firebase to a carer's mobile application and are shown on a web dashboard.

The central value of the solution is that the warning signs are interpreted together and locally. The belt's posture changes how the glove's heart rate is read; the outputs of five screening engines are merged into a single Master Risk Score; and one alert channel serves every emergency, whether it comes from a model, a fall or an SOS gesture. Detection does not wait for the internet, and raw ECG and motion data stay on the hub. The intended end-to-end outcome is that a carer is told, within seconds and with a clear reason, that the person they look after needs attention.

3.2 High level architecture

Figure 3.1 shows the architecture of the complete solution. It has three deployment zones. The wearable zone contains the glove, built on an ESP32 microcontroller, and the belt, built on an ESP32-S3. The hub zone is a computer on the local network running a Python Flask backend, which exposes the /api/telemetry endpoint, hosts the screening engines of components C1, C3 and C4, the shared sensor fusion module, the correlation engine, the Master Risk Score, the alert manager and the web dashboard. The cloud zone contains the Firebase Realtime Database and the carer mobile

application. The colours show which component owns each part; grey parts were built jointly.

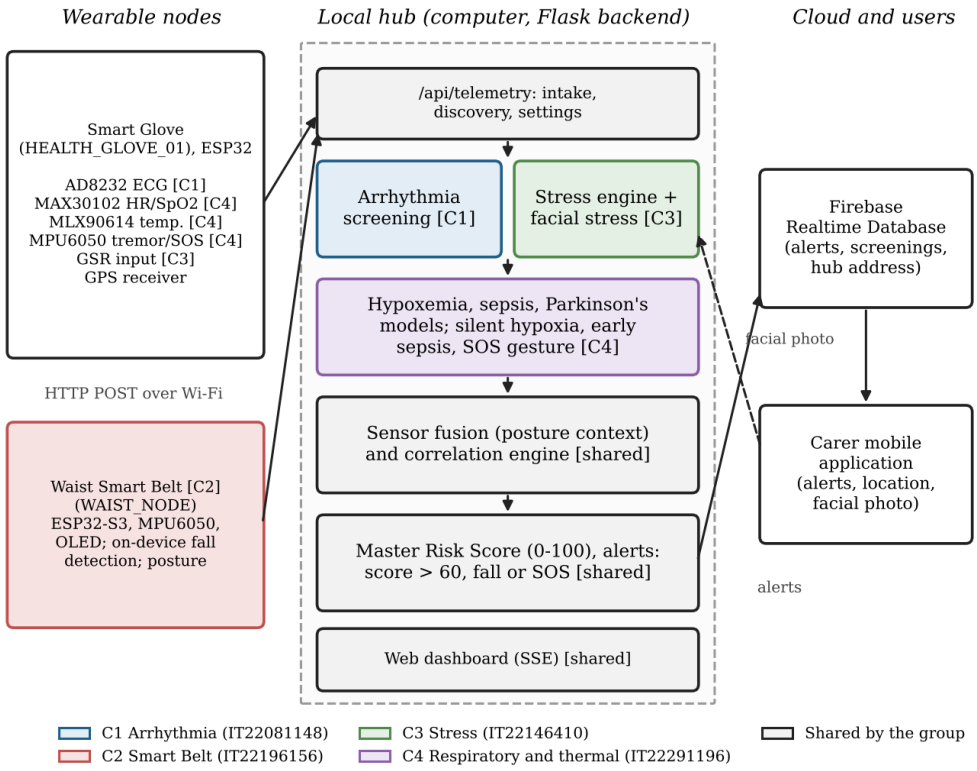


Figure 3.1: High-level architecture of the integrated SmartGlove solution

Source: Group, based on the project source code and the individual theses.

3.3 Component 1 summary

Table 3.1 summarises Component 1 in the common format used for all four components.

Table 3.1: Summary of Component 1, arrhythmia screening

Aspect	Description
Owner	Abeysinghe S.A.M.M. (IT22081148)
Subproblem	SP1: heart rhythm abnormalities go unnoticed
Input	Single-lead ECG from the AD8232 on the glove at 250 Hz; heart rate; belt posture

Aspect	Description
Core method	R-peak detection and sixteen beat features on the latest eight seconds of ECG; 240-tree Random Forest classifying N, SVEB, VEB, F and Q beats; sustained bradycardia (< 45 BPM for 10 readings) and tachycardia (> 140 BPM) rules; posture-aware suppression of activity-explained high heart rate
Output	Diagnosis, severity (0 to 2) and abnormal likelihood; LOW_HR and HIGH_HR rule alerts
Contribution to the whole	Cardiac screening input to the Master Risk Score; first user of belt posture as context

Source: Individual thesis [12].

The arrhythmia component turns the raw ECG into a beat-by-beat screening result on the hub, so that ectopic beats are noticed while they happen [12]. Its detailed dataset analysis, classifier comparison and simulation tests of the live R-peak detector are reported in the individual thesis.

3.4 Component 2 summary

Table 3.2 summarises Component 2 in the common format used for all four components.

Table 3.2: Summary of Component 2, waist-worn Smart Belt

Aspect	Description
Owner	Lakshan N K T (IT22196156)
Subproblem	SP2: falls are not reported when the person cannot call for help
Input	MPU6050 acceleration and angular velocity at the waist, sampled every 100 ms with a 21 Hz hardware low-pass filter
Core method	Signal magnitude vector and trunk tilt; two-stage rule (impact above 24.0 m/s² followed within 1.5 s by tilt above 55°); 15 s alert latch; posture from tilt and acceleration variance
Output	fall_detected flag and posture (Standing, Sitting, Walking, Running, Lying), sent to the hub at 2 Hz and immediately on a new fall
Contribution to the whole	The most urgent emergency trigger, and the activity context used by the hub's sensor fusion

Source: Individual thesis [13].

The belt makes the fall decision on the wearable itself, so it does not depend on the hub or the network to detect a fall, only to deliver the alert [13]. Its posture output is the main piece of shared context in the system.

3.5 Component 3 summary

Table 3.3 summarises Component 3 in the common format used for all four components.

Table 3.3: Summary of Component 3, Stress Monitoring System

Aspect	Description
Owner	E.A.P. Peshala (IT22146410)
Subproblem	SP3: stress and well-being are not observed
Input	GSR sampled on the glove at 50 Hz; heart rate and finger temperature; a face photograph from the mobile application
Core method	Part A: Random Forest on statistical heart-rate and temperature features over 120 s windows, blended with a GSR score and falling back to vital signs only when electrodes detach. Part B: Haar cascade face detection and a fine-tuned MobileNetV2 classifier
Output	Stress likelihood on the hub (screenings.stress); stressed or not stressed result for the photograph, returned through Firebase
Contribution to the whole	Psychological well-being dimension; stress likelihood enters the Master Risk Score

Source: Individual thesis [14].

The stress component adds a dimension that the cardiorespiratory engines do not cover, using sensors already on the glove and the carer's phone camera [14].

3.6 Component 4 summary

Table 3.4 summarises Component 4 in the common format used for all four components.

Table 3.4: Summary of Component 4, Respiratory and Thermal Warning System

Aspect	Description
Owner	Dissanayaka D.M.S.K. (IT22291196)
Subproblem	SP4: respiratory, thermal and motor decline goes unnoticed

Aspect	Description
Input	Heart rate and SpO2 from the MAX30102, skin temperature from the MLX90614, and glove motion from the MPU6050 at 50 Hz
Core method	Hypoxemia screening (logistic regression, SpO2 below 92%); sepsis screening (320-tree Random Forest on heart rate, SpO2 and temperature); Parkinson's tremor screening (320-tree Random Forest on 22 motion and frequency features); rule-based silent hypoxia and early sepsis detectors; SOS shake gesture detection
Output	Likelihoods for hypoxemia, sepsis and tremor; rule alerts; SOS trigger
Contribution to the whole	Respiratory, infection and motor screening inputs to the risk score, and the user-initiated emergency trigger

Source: Individual thesis [15] and the project's model documentation.

This component covers the largest number of conditions, which reflects the number of sensors on the glove that relate to breathing, temperature and movement [15].

3.7 End to end user and data flow

Figure 3.2 shows how data move through the complete solution. The glove and belt sample their sensors continuously. Their messages reach the hub's /api/telemetry endpoint over HTTP, where the backend stores the latest state of each device. With each update, the screening engines of C1, C3 and C4 run on the newest data and report a likelihood once they have enough data to be "ready". The sensor fusion module then compares the glove's vital signs with the belt's posture, the heart-rate rules are applied, and activity-explained heart-rate alerts are removed. The Master Risk Score is calculated from the highest likelihood among the ready engines and the correlation engine's measure of deviation from the user's own baseline. When the score exceeds 60, the alert is written to Firebase and delivered to the carer's phone, and the dashboard shows it immediately.

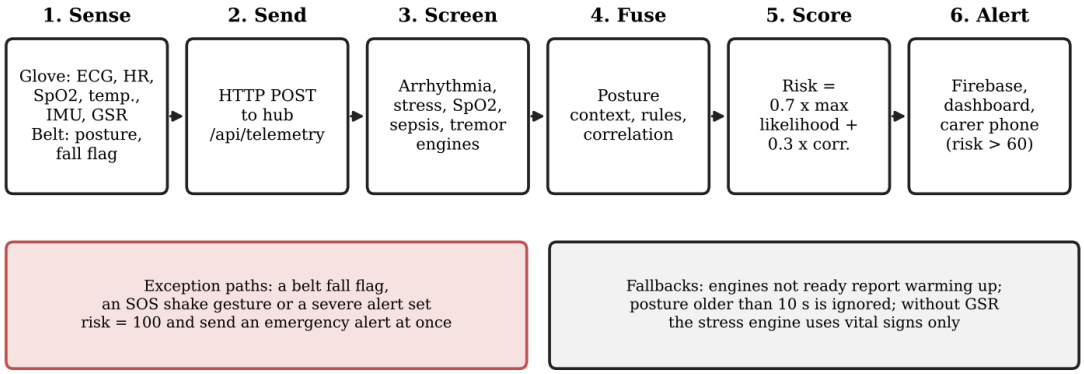


Figure 3.2: End-to-end data flow through the integrated solution, with exception and fallback paths

Source: Group.

Three events bypass the score: a fall flag from the belt (C2), an SOS shake gesture detected from the glove's motion (C4), and any alert marked as severe. Each sets the risk to 100 and raises an emergency at once. The stress component's facial check follows a separate request path: the carer takes a photograph in the mobile application, it is passed through Firebase to the hub, classified, and the result is written back to the application.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

This chapter explains how the four components were brought together: the integration strategy, the interfaces and shared data, the non-functional requirements common to the whole system, the integrated workflow and its exception handling, and the division of work and dependencies within the team.

4.1 Integration strategy

The team chose a hub-centred integration strategy. Instead of connecting components to each other directly, every wearable sends its data to the hub, and every screening engine reads from and writes to the hub's shared state. This kept the interfaces between members few and simple: one telemetry endpoint, one screening output format, one alert format and one set of Firebase paths.

Integration was interface-first. The JSON fields of the glove and belt messages and the dictionary returned by each screening engine were agreed early, so that each member could develop and test a component against the agreed format before the others were finished. Two features of the backend reduced the risk of waiting for hardware. A simulation mode (simulator.py) injects fast-forwarded clinical data into the telemetry stream, and the glove has a simulated vital-sign setting (syncMode), read from the hub's /api/settings endpoint, so engines could be exercised without a person wearing the devices [14]. Components were then connected in stages: the belt and glove to the hub, each screening engine to the shared state, the fusion module and risk score, and finally Firebase, the dashboard and the mobile application.

Interface ownership followed the components. The belt message belonged to C2, the screening outputs to their engine owners, and the shared hub services, fusion module, correlation engine, Master Risk Score, dashboard and mobile application were owned jointly. The main integration risks and how they were handled are listed in Table 4.1.

Table 4.1: Main integration risks and how they were managed

Risk	Effect	Mitigation
Hardware not ready while models are developed	Engines cannot be tested	Simulation mode on the hub and syncMode on the glove

Risk	Effect	Mitigation
Hub address changes on the network	Wearables lose the hub	Discovery by UDP broadcast, then hub address in Firebase, then a fixed fallback address
Posture data arrive late or stop	Wrong suppression of heart-rate alerts	Fusion confidence falls with the age of the posture; posture older than 10 s is ignored
An engine has too little data at start-up	Unreliable early outputs	Engines report "warming up" and are excluded from the risk score until ready
One condition diluted by healthy engines	Missed alert	Risk uses the maximum likelihood, not the average

Source: Group, from the project source code and the individual theses.

4.2 Interfaces data contracts and shared models

Table 4.2 summarises the interfaces between the components. The detailed field-level specification is given in Appendix A.

Table 4.2: Interfaces between the components and the shared services

Interface	Direction	Format	Owner
/api/telemetry	Glove and belt to hub	HTTP POST, JSON; device_id identifies the node (HEALTH_GLOVE_01, WAIST_NODE)	Shared (belt message: C2)
/api/settings	Hub to glove	HTTP GET, JSON (syncMode)	Shared
Hub discovery	Wearables to hub	UDP broadcast; Firebase hub address; fixed fallback	Shared
Screening output	Each engine to hub state	Dictionary: diagnosis, severity, likelihood_percent, status (ready or warming_up)	C1, C3, C4
Rule and fall alerts	Engines and rules to alert manager	Dictionary: diagnosis, severity, trigger, value	All
Processed state	Hub to Firebase	/processed/HEALTH_GLOVE_01 with screenings, alerts and vitals	Shared
Facial stress request	Mobile app to hub through Firebase	Base64 image, pending flag, device identifier; result with class, confidence and bounding box	C3
/api/facial-stress	Hub endpoint	JSON	C3
Dashboard stream	Hub to browser	Server-Sent Events	Shared

Source: Group, from the project source code and the individual theses.

Two shared models tie the components together. The first is the per-device state on the hub, which keeps the latest message and its arrival time for each node. Fusion uses the arrival time of the belt's posture to judge whether it is still fresh. The second is the screening output format. Because every engine returns the same four fields, the risk score can treat them uniformly: it reads likelihood_percent from every engine whose status is ready. Severity uses a shared scale: 0 is normal, 1 is a warning, 2 is a serious finding and 3 is reserved for falls. Each component validates its own inputs. For example, the arrhythmia engine returns no prediction when the ECG window is too short or flat, and the belt marks posture as UNKNOWN when it cannot classify it.

4.3 Shared nonfunctional requirements

Table 4.3 lists the non-functional requirements that apply to the whole system and how each was addressed.

Table 4.3: Shared non-functional requirements

Requirement	Target	How it is addressed
Performance	Each hub model well within the 500 ms update interval; alert within seconds	All hub models run in under 4 ms on a CPU; immediate transmission of a new fall
Reliability	Detection continues when the internet is lost	Falls detected on the belt; models run on the local hub
Usability	No action needed from the older user	Passive sensing; alerts go to carers; SOS gesture as an optional manual trigger
Privacy	Raw ECG and motion stay local	Only processed state and alerts are sent to Firebase [17]
Security	Data protected in transit	Not met in the prototype: plain HTTP on the local network (Section 6.6)
Explainability	Alerts explainable in physical terms	Threshold rules and tree-based models with inspectable features
Maintainability	Members can change their component independently	One engine module per component; shared output format
Scalability	Several users per hub	Device identifiers in every message; per-user Firebase paths needed (Section 7.4)

Source: Group.

4.4 Integrated workflow and exception handling

In normal operation, the glove and belt send telemetry about twice per second, the engines update, the risk score stays low and the dashboard shows the current state. Table 4.4 describes how the integrated system behaves in the main failure and exception cases.

Table 4.4: Exception handling in the integrated workflow

Situation	System behaviour
Fall detected by the belt	Immediate message to the hub; risk set to 100; FALL DETECTED alert (severity 3) with location to Firebase; belt shows an emergency screen
SOS shake gesture on the glove	Risk set to 100; emergency alert sent
High heart rate while running or walking	HIGH_HR alert suppressed if posture is fresh and explains the rate; arrhythmia beat alerts are kept
Posture data stale (> 10 s)	Fusion confidence drops to 0.5 or below; heart-rate alerts are no longer suppressed
ECG electrode loose or signal flat	Arrhythmia engine returns no features; median imputation covers missing values
GSR electrodes detached	Stress engine falls back to the vital-sign-only model
Wi-Fi lost on the belt	Belt keeps detecting and displaying falls; reconnects every 5 s and resends
Hub address changed	Belt repeats discovery after three failed transmissions
Alert dismissed by a nurse	System returns to normal monitoring

Source: Group, from the project source code and the individual theses.

4.5 Team contribution and dependency matrix

Table 4.5 maps each member to their component, research and development work, integration responsibilities and test evidence. Table 4.6 shows which components depend on which. The full contribution declaration agreed by all members is given in Appendix B.

Table 4.5: Team contribution matrix

Member	Component and research	Development	Integration work	Test evidence
Abeysinghe S.A.M.M. (IT22081148)	C1 Arrhythmia: dataset analysis, classifier comparison, R-peak detector study	ECG feature extraction, Random Forest pipeline, heart-rate rules, posture-aware suppression	Screening output to risk score; use of belt posture	Held-out evaluation; 17 unit tests; simulation tests
Lakshan N K T (IT22196156)	C2 Smart Belt: fall detection and posture literature	Belt hardware and firmware, two-stage fall rule, posture classifier, OLED	Belt telemetry and discovery; backend fall handling	120 fall and activity trials; 24 end-to-end tests
E.A.P. Peshala (IT22146410)	C3 Stress: physiological and facial stress literature	GSR acquisition, stress engine, facial pipeline, /api/facial-stress	Glove GSR field; Firebase facial exchange; stress on dashboard and app	Subject-wise evaluation; functional tests; latency
Dissanayaka D.M.S.K. (IT22291196)	C4 Respiratory and thermal: oximetry, sepsis and tremor literature	Hypoxemia, sepsis and tremor models; silent hypoxia and early sepsis rules; SOS gesture	Glove vital-sign fields; SOS emergency path	Held-out model evaluation
All members	Common problem and architecture	Flask hub, sensor fusion, correlation engine, Master Risk Score, dashboard, mobile application	Telemetry format, Firebase paths, alert workflow	Integration testing (Section 6.2)

Source: Group.

Table 4.6: Dependency matrix (row depends on column)

Depends on	C1 Arrhythmia	C2 Belt	C3 Stress	C4 Respiratory	Shared hub and alerts
C1 Arrhythmia	-	Posture for suppression	-	Glove heart rate (MAX30102)	Telemetry, risk score, Firebase
C2 Belt	-	-	-	-	Hub endpoint, alert delivery
C3 Stress	-	-	-	Glove heart rate and temperature	Telemetry, Firebase, app, risk score
C4 Respiratory	-	Posture for context	-	-	Telemetry, risk score, alerts
Shared hub and alerts	Likelihood	Fall flag, posture	Likelihood	Likelihoods, SOS	-

Source: Group.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

This chapter describes the technology shared by the team, the development and integration process, how the integrated system is configured and run, and the evidence of how the team carried out the project.

5.1 Common technology stack and infrastructure

Table 5.1 lists the shared technology. The main choices were made for three reasons: low cost, the ability to run every model on an ordinary computer without a graphics card, and a single language (Python) on the hub so that all members could read and test each other's engines.

Table 5.1: Common technology stack

Layer	Technology	Used by	Reason for choice
Wearable microcontrollers	ESP32 (glove), ESP32-S3 SuperMini (belt), Arduino framework	All	Low cost, built-in Wi-Fi, enough processing for on-device rules
Sensors	AD8232, MAX30102, MLX90614, MPU6050 (x2), GSR module, GPS receiver	C1 to C4	Widely available low-cost modules
Firmware tools	Arduino IDE 2.3.2, ESP32 core 3.0.0, Adafruit sensor and display libraries	C2, glove firmware	Mature libraries; fast development
Transport	Wi-Fi, HTTP POST with JSON; UDP broadcast for discovery	All	Simple, debuggable, works on any local network
Hub	Local computer, Python 3.11, Flask 3.0 or later	All	One backend for all engines; runs on existing hardware
Machine learning	scikit-learn 1.3 or later, joblib, NumPy, SciPy; TensorFlow/Keras and OpenCV for the facial model	C1, C3, C4	CPU-only inference in milliseconds
Cloud	Firebase Realtime Database (firebase-admin 6.5 or later)	All	Real-time sync to the mobile app; free tier sufficient for a prototype
User interfaces	Flask templates with Server-Sent Events; carer mobile application	All	Live dashboard on the hub; alerts and photos on the phone

Source: Group, from the project requirements file and the individual theses.

The source code is organised by component inside one repository. The backend (backend/main.py) handles telemetry, rules, the risk score and alerts. Each screening engine has its own module under backend/AI_models, with the shared modules sensor_fusion.py and correlation_engine.py and the rule engines beside them. Each trained model is saved with its metrics file in the module's saved_model folder, and the firmware for each wearable (glove.ino, belt.ino) is kept in its own folder.

5.2 Development and integration process

Development ran in parallel streams, one per component, connected by the agreed interfaces. Each member developed and tested their component in isolation first: the belt against its serial output and bench tests, and the screening engines against held-out data and the hub's simulation mode. Integration then followed the staged sequence described in Section 4.1. Problems found during integration were fixed by the owner of the interface concerned, or jointly where they affected the shared hub. Examples include stale posture data, which was handled by the fusion confidence, and the hub address changing between sessions, which was handled by the discovery sequence.

5.3 Deployment and operational configuration

The integrated system is deployed in four steps.

1. **Hub.** The Python dependencies are installed from the requirements file on the hub computer, the Firebase service account key is placed in the backend folder, and the Flask backend is started. On start-up it loads every saved model, and engines report "warming up" until they have enough data.
2. **Wearables.** The glove and belt firmware are flashed with the Wi-Fi credentials. On power-up each device connects to Wi-Fi, discovers the hub and begins sending telemetry; the belt's OLED shows whether Wi-Fi and the server are connected.
3. **Cloud and app.** The hub publishes its address and processed state to Firebase. The carer signs in to the mobile application, which reads alerts and screenings from Firebase.

4. **Operation.** The web dashboard on the hub shows live vitals, screenings and alerts. A nurse or carer can dismiss an alert, which returns the system to monitoring.

Monitoring of the system itself relies on the dashboard, the backend log and the wearables' serial diagnostics. The trained models and their metrics files are versioned with the source code, so the hub can be rebuilt by reinstalling the dependencies and copying the repository. Firebase keeps the most recent state and alerts. A formal backup and restore procedure for alert history was not implemented and is noted as future work.

5.4 Evidence of team execution

The team's work is recorded in four places: task plans in MS Planner, meetings and discussions in Microsoft Teams, the commit history of the shared Git repository, and each member's research diary. Appendix B summarises the agreed division of work.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

This chapter sets out how the integrated solution was evaluated, reports the integration and end-to-end tests, summarises the results of all four components, describes stakeholder validation, maps the evidence to the group objectives, and discusses the limitations of the system as a whole.

6.1 Evaluation framework and success criteria

The group evaluation was organised around four questions:

- **EQ1.** Does each component detect what it is meant to detect, on data or trials it has not seen before?
- **EQ2.** Do the components run fast enough on the chosen hardware to support real-time alerts?
- **EQ3.** Does an event detected anywhere in the system reach a carer's phone reliably and quickly?
- **EQ4.** Does the shared context, the belt's posture, change the system's decisions in the intended way?

Table 6.1 lists the acceptance criteria, metrics and evidence agreed for each question. The component evaluations used held-out data split by subject or record wherever possible, so that no person's data appeared in both training and testing, and the fall trials used healthy adult volunteers performing simulated falls onto a mattress [13].

Table 6.1: Group evaluation questions, metrics and acceptance criteria

Question	Metric	Acceptance criterion	Evidence
EQ1 Detection	Accuracy, balanced accuracy, recall; fall sensitivity and specificity	Recall of the critical class reported; fall sensitivity and specificity above 90%	Component held-out tests; belt trials
EQ2 Speed	Inference time per prediction	Under 10 ms per hub model on a CPU	Component timing measurements
EQ3 Alert delivery	Share of alerts delivered; time from event to phone	At least 95% delivered; mean under 3 s	End-to-end alert tests
EQ4 Shared context	Correct suppression and non-suppression of heart-rate alerts	All defined scenarios behave as designed	Unit and scenario tests of the fusion logic

Source: Group.

6.2 Integration and end to end testing

Table 6.2 summarises the integration and end-to-end tests carried out on the complete system.

Table 6.2: Integration and end-to-end test results

Test	Scope	Result	Source
Belt telemetry reaches the hub	C2 to hub	Messages received; fall produces emergency with risk 100 and severity 3	[13]
Fall alert end to end	Belt, hub, Firebase, phone	23 of 24 alerts delivered (95.8%); mean 1.8 s from fall flag to notification	[13]
Alert dismissal	Phone or dashboard to hub	23 of 23 cleared correctly	[13]
Belt resilience	Wi-Fi off, hub address changed	Sensing continued; reconnection and rediscovery worked	[13]
Heart-rate rules and posture fusion	Hub logic used by C1 and C2	17 of 17 unit tests passed, including running (suppressed), lying (kept) and stale posture (kept)	[12]
Arrhythmia alert to Firebase and dashboard; exercise scenario	C1, hub, Firebase, dashboard	Recorded as passed in integration testing	[12]
Facial stress request	App, Firebase, hub, app	Processed end to end; about 150 ms processing on a CPU	[14]
Stress engine integration	Glove GSR, hub, dashboard, Firebase	Functional tests reported in the thesis	[14]

Source: Individual theses.

The strongest end-to-end evidence concerns the fall path, which was tested 24 times through the full chain from the belt to the phone. The other alert paths share the same risk score, Firebase write and phone notification, so the 1.8 s delivery time is the best available estimate of the delivery delay for every emergency. Security, load and formal usability testing of the integrated system were not carried out.

6.3 Overall results

Table 6.3 and Figure 6.1 summarise the main result of each component. Every figure comes from the component's own held-out evaluation, and the individual theses give the full experiments.

Table 6.3: Summary of component results

Component	Task	Evaluation data	Main result	Time per prediction
C1	Arrhythmia, 5 beat classes	10,000 beats from 20 held-out synthetic records	Accuracy 80.7%, balanced accuracy 68.9%; VEB recall 73.3%, SVEB recall 36.9%	3.4 ms
C2	Fall detection	48 simulated falls and 72 daily activities	Sensitivity 95.8%, specificity 97.2%	On device, 0.8 s mean detection after impact
C2	Posture	120 posture trials	96.7% correct	On device
C3	Stress from vital signs and GSR	2,280 windows from 38 held-out subjects	Accuracy 96.6%, balanced accuracy 96.2%, stressed recall 95.4%	2.9 ms
C3	Facial stress	37 validation images	Best validation accuracy 86.5% (signs of overfitting)	About 150 ms
C4	Hypoxemia (SpO2 below 92%)	100 monitoring profiles, held-out readings	Accuracy 98.9%; 831 of 831 low-oxygen readings detected	Under 0.1 ms
C4	Sepsis	5,000 hourly windows from 100 held-out patients	Accuracy 95.0%, balanced accuracy 89.6%, F1 84.1% for sepsis	3.1 ms
C4	Parkinson's tremor	1,800 recordings from held-out subjects	Accuracy 92.4%; F1 92.0% for Parkinson's	3.8 ms

Source: Individual theses [12], [13], [14] and the project's model documentation for C4.

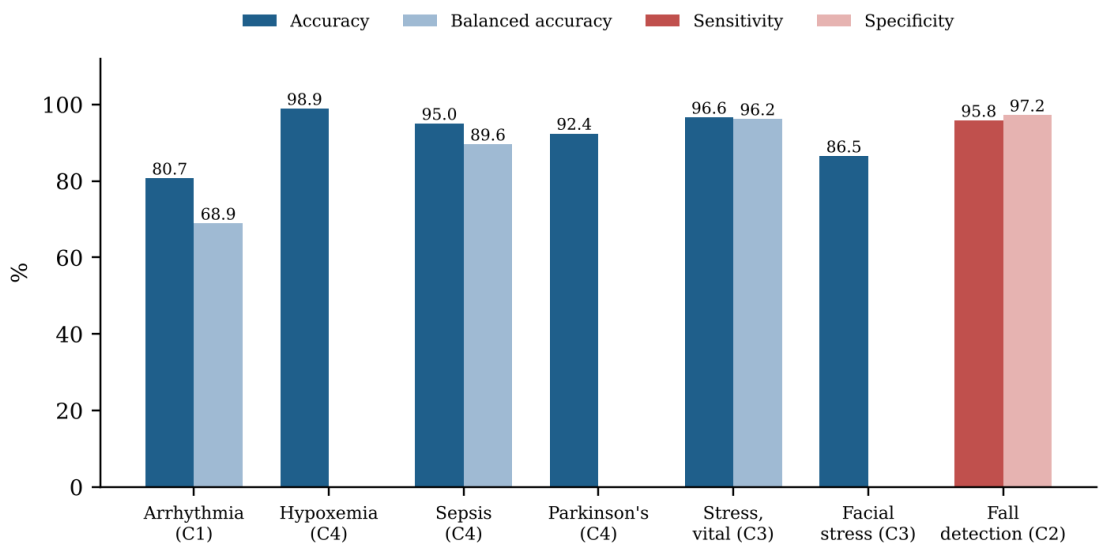


Figure 6.1: Main detection results of the components (accuracy and balanced accuracy for screening models; sensitivity and specificity for fall detection)

Source: Group, from Table 6.3.

Taken together, the results answer EQ1 and EQ2 positively at the level of technical feasibility. The fall detector met its criterion on physical trials, and every hub model ran in a few milliseconds, which is far below the half-second interval between telemetry updates. The screening results differ widely in difficulty. Hypoxemia screening is almost perfect, partly because the label is defined directly from the SpO2 value the model reads (Section 6.6). Arrhythmia screening is the hardest task, and its balanced accuracy shows that supraventricular and fusion beats are often missed [12]. EQ3 is answered for the fall path (95.8% of alerts delivered, mean 1.8 s), and EQ4 is answered by the passing unit and scenario tests of the posture fusion.

6.4 Stakeholder and user validation

No formal validation of the complete system with older adults, carers or clinicians was carried out within the project period. The volunteers who took part in the fall and activity trials were healthy adults, and they tested the belt rather than the whole system [13]. Feedback from the supervisors at the progress reviews shaped several design decisions, such as making the fall decision on the belt and suppressing only heart-rate

alerts during activity, but this is not user validation. Validation with the intended users is the most important item of future work (Section 8.3).

6.5 Achievement of common objectives

Table 6.4 maps the evidence to each group objective.

Table 6.4: Achievement of the group objectives

Objective	Evidence	Status
GO1 Acquire and stream all signals to the hub	Glove and belt telemetry received by the hub; component sensor tests	Achieved
GO2 Real-time screening models on a CPU	Held-out results for five model types; all under 4 ms except facial stress (about 150 ms)	Achieved, with dataset caveats
GO3 On-device fall detection and posture	95.8% sensitivity, 97.2% specificity; posture 96.7%	Achieved under controlled conditions
GO4 Posture-aware suppression of heart-rate alerts	17 unit tests passed; exercise scenario passed	Achieved
GO5 One risk score and alerts with location	Fall alerts 23 of 24 delivered in 1.8 s mean; risk score implemented for all engines	Partly achieved: end-to-end timing measured only for the fall path
GO6 Local, private and explainable decisions	Models and rules on the hub; feature-based and threshold-based outputs	Partly achieved: communication not encrypted

Source: Group.

6.6 Integrated limitations and threats

Several limitations apply to the system as a whole, beyond those of each component.

- **Training data.** The arrhythmia, hypoxemia, sepsis and Parkinson's models were trained on public Kaggle datasets from the same publisher. The arrhythmia dataset was confirmed to be synthetic [12], and the others should be treated the same way until their origin is verified. The origin of the stress dataset is not documented [14]. All accuracy figures therefore show technical feasibility, not clinical accuracy.

- **Easy targets.** The hypoxemia label is SpO2 below 92%, and SpO2 is also the model's main input, so near-perfect results mostly show that the model learned the threshold.
- **Controlled trials.** Falls were simulated by healthy adults onto a mattress, and real falls by older people or people with Parkinson's disease differ [13].
- **Live signal path.** For the arrhythmia component, simulation showed that the live R-peak detector double-counts beats at resting heart rates, so the deployed pipeline would perform worse than the dataset figure suggests until the proposed fix is adopted [12].
- **Single point of failure.** All screening, and the delivery of all alerts, depend on the hub computer being on. Alerts to carers also depend on internet access for Firebase.
- **Security.** Telemetry travels over plain HTTP without authentication, and Wi-Fi credentials are stored in the firmware, so an attacker on the local network could read or forge data, including a false fall [13].
- **Interaction between components.** The risk score takes the highest likelihood from engines of very different reliability, so a weak engine can raise or mask risk. Thresholds such as 60 for the risk score were set by judgement, not tuned on recorded scenarios.
- **Evidence for integration.** End-to-end timing was measured for the fall path only, and no multi-hour run of the full system with all engines active has been reported.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

This chapter considers how SmartGlove could become a product: the value it creates, its position against alternatives, a possible business model, the barriers to adoption and the compliance requirements, and its plausible contribution to the Sustainable Development Goals. It is an analysis of potential, and no market sizes or prices are stated without a verified source.

7.1 Value proposition and beneficiaries

For older adults, SmartGlove offers continuous monitoring that needs no action from them, including help after a fall when they cannot call. For families and home carers, it offers one alert stream that says what happened and where, instead of several apps or none. For care homes, it offers a single dashboard covering cardiac, respiratory, stress and fall risks for each resident, which keeps working when the internet is unreliable. For clinicians, it offers screening flags based on standard measures, with decisions that can be explained. The value of the integrated product is greater than the sum of the four components, because the components share sensors, context and one alert channel: the belt makes the glove's heart-rate alerts more trustworthy, and a single kit replaces several separate devices.

7.2 Market alternatives and differentiation

The primary target segments are families caring for an older relative at home, and private care homes and home-care services. People with Parkinson's disease, who fall often, are a focused secondary segment. Table 7.1 compares SmartGlove with the alternatives available to these segments.

Table 7.1: Comparison of SmartGlove with alternative solutions

Feature	Smartwatch with health and fall features	Personal emergency pendant	Holter or ECG patch	SmartGlove
Signals monitored	Heart rate, ECG on demand, falls	Button, some fall detection	ECG only	ECG, HR, SpO2, temperature, motion, GSR, falls, posture

Feature	Smartwatch with health and fall features	Personal emergency pendant	Holter or ECG patch	SmartGlove
Real-time alerts	Some	Yes (falls, button)	No	Yes, all conditions
Shares context between sensors	No	No	No	Yes (posture)
Works without internet for detection	Partly	Partly	Yes (records only)	Yes (local hub)
Open, explainable decisions	No	No	No	Yes
Clinically validated	Yes, for specific features [18]	Yes, for falls	Yes [10]	No (prototype)

Source: Group, based on the individual theses and the cited works.

SmartGlove's defensible differences are the integration of several risks in one low-cost kit, the use of posture as context, local decision-making, and transparent rules and models. Its main weakness against all alternatives is the lack of clinical validation, which is the first barrier to adoption.

7.3 Business model and feasibility

The proposed model, consistent across the four theses, is to sell or lease the complete kit (glove, belt and hub software) rather than the components separately, with an optional subscription for the cloud alert service and mobile application [13], [14]. Two channels are proposed: direct sales to families, and institutional licences for care homes and home-care providers, where one hub can serve several residents. Partners would include private hospitals offering home-care services and Parkinson's support groups.

The main costs are the hardware, the cloud service and support. The hardware uses low-cost modules; the belt's bill of materials is estimated at about USD 25 [13], and the other components add mainly the ECG, pulse oximeter, temperature and GSR modules to the glove. The hub runs on an ordinary computer, and the hub models are small and fast enough for many users per hub [14]. Cloud costs at prototype scale fit within the free tier of Firebase.

Technical feasibility is supported by the working integrated prototype and the component results in Chapter 6. Operational feasibility depends on setting up Wi-Fi and the hub in each home, on users wearing two devices, and on a support process for charging, electrode replacement and alert handling.

7.4 Adoption intellectual property and compliance

Adoption barriers. Older users may find two devices and ECG electrodes uncomfortable for daily wear. Carers will stop trusting the system if false alarms are frequent. Institutions will want evidence of clinical benefit before paying.

Intellectual property and licensing. The methods used, such as threshold fall detection, Random Forests, logistic regression and MobileNetV2, are published and implemented in open-source libraries, so the value of the product lies in the integration, the implementation and future validation rather than in patentable algorithms [13], [14]. Library licences must be respected: the Arduino core for the ESP32 is distributed under the LGPL 2.1 and the Adafruit libraries under BSD licences [13]. The arrhythmia training dataset is distributed under GPL 2, which must be reviewed before a model trained on it is used commercially [12]. Ownership of the code within the group and with SLIIT should be agreed in writing.

Data protection and safety. Health, location and facial data are sensitive personal data under Sri Lanka's Personal Data Protection Act, No. 9 of 2022 [17]. Deployment would require user consent, per-user access control on Firebase, encrypted communication, deletion of facial images after processing and a defined retention period. To scale to many users, each glove and belt also needs a unique identifier, because the current firmware uses fixed identifiers.

Regulation. Software that screens for arrhythmia, hypoxemia or sepsis would be regulated as software as a medical device [19], and in Sri Lanka medical devices fall under the National Medicines Regulatory Authority Act, No. 5 of 2015 [20]. A product would need risk management according to ISO 14971 [21] and a software life cycle according to IEC 62304 [22]. The prototype was not developed under these standards, and no claim of compliance is made.

7.5 Sustainable Development Goal contribution

The plausible contribution to SDG 3 is earlier detection of deterioration and faster help after falls in older adults, which relates to target 3.4 on noncommunicable diseases and mental well-being [11]. The contribution to SDG 10 is lower-cost monitoring that does not need reliable internet for detection. Neither contribution has been measured.

Possible negative effects must also be considered. False reassurance could lead families to check on a relative less often, false alarms could cause anxiety and alert fatigue, and continuous monitoring raises privacy concerns, especially for facial images and location. To sustain the benefits, the system would need clinical validation, clear communication that it is a screening aid and not a diagnosis, strong data protection, and hardware that can be repaired and updated by firmware rather than replaced.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The common problem was how to watch several warning signs in an older adult continuously at home, interpret them together and turn them into timely, trustworthy alerts, at low cost and without depending on the internet for the decision. The integrated evidence shows that this is technically achievable with inexpensive hardware. SmartGlove collects cardiac, respiratory, thermal, motion and stress signals from a glove and a belt, makes every detection decision on the wearables or a local hub, interprets the heart rate using the belt's posture, merges five screening engines into one risk score, and delivers emergencies to a carer's phone. Fall alerts, the most urgent path, were detected in 46 of 48 simulated falls and reached the phone in 23 of 24 end-to-end tests with a mean delay of 1.8 s. All hub models ran in a few milliseconds on a CPU.

The evidence does not show that the system is clinically accurate. Most models were trained on synthetic or undocumented data, falls were simulated by healthy adults, and no older adults or clinicians evaluated the complete system. The answer to the common problem is therefore a qualified yes: the integrated approach works as a system, and its value now depends on validation with real users and real data.

8.2 Principal integrated contribution

Beyond the four components, the team produced three things of value. First, a working architecture in which wearables, screening engines and alerting are connected through one hub with a small set of agreed interfaces, so that new engines can be added without changing the others. Second, a demonstration that sharing context between sensors is practical and useful: the belt's posture, produced for fall detection, is reused to make heart-rate alerts more trustworthy. Third, a single, explainable alert pipeline in which a model likelihood, a fall, an SOS gesture and a rule alert all lead to the same kind of emergency message with location. The team also learned that testing the live signal path, and not only the dataset, is essential: one component found a serious weakness in its live R-peak detector that the dataset accuracy did not reveal.

8.3 Future directions

The following work is recommended, in order of priority.

1. **Validate with the intended users.** Conduct a supervised pilot with older adults and carers, with ethical approval, to measure comfort, false alarm rates over days of wear, and carer response.
2. **Retrain and test on real data.** Replace synthetic training data with clinical or volunteer recordings from the actual hardware, starting with ECG and SpO2, and report results on people not used for training.
3. **Secure the communication.** Add encryption and device authentication between the wearables and the hub, remove credentials from the firmware, and apply per-user access rules in Firebase.
4. **Measure every alert path end to end.** Repeat the end-to-end tests for arrhythmia, hypoxemia, sepsis, stress and SOS alerts, and run the full system for several hours with all engines active.
5. **Tune the shared decision layer.** Set the risk score weights and threshold on recorded scenarios, and weight engines by their measured reliability.
6. **Reduce dependence on the hub.** Move the most critical checks, such as falls and severe heart-rate or SpO2 limits, onto the wearables or a phone so that an alert can still be raised if the hub is off.
7. **Prepare for regulation.** Begin risk management and software life-cycle documentation in line with ISO 14971 and IEC 62304 if the product is to be developed further.

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APPENDICES

Appendix A Integrated interface specification

This appendix specifies the interfaces summarised in Section 4.2. Table A.1 lists the hub endpoints, Tables A.2 and A.3 the two telemetry messages, Tables A.4 and A.5 the shared screening and alert formats, and Table A.6 the Firebase paths. Field names are taken from the project source code and the individual theses.

Table A.1: Hub endpoints

Endpoint	Method	Caller	Purpose
/api/telemetry	POST	Glove, belt	Receives JSON telemetry; also answers belt discovery requests
/api/settings	GET	Glove	Returns settings such as syncMode (simulated vital signs)
/api/facial-stress	POST	Hub listener / app path	Classifies a face image as stressed or not stressed
Dashboard stream	GET (SSE)	Browser	Pushes processed state and alerts to the web dashboard

Source: Project source code and individual theses.

Table A.2: Belt telemetry message (device_id WAIST_NODE)

Field	Type	Description
device_id	string	Fixed identifier "WAIST_NODE"
node_type	string	Fixed value "waist"
timestamp	number	Seconds since belt start-up (millisecond resolution)
ax, ay, az	number	Acceleration in m/s²
gx, gy, gz	number	Angular velocity in °/s
tilt_degrees	number	Trunk tilt angle in degrees
posture	string	STANDING, SITTING, LYING, WALKING, RUNNING or UNKNOWN
activity_magnitude	number	Variance of the signal magnitude vector over 20 samples
fall_detected	boolean	True while a confirmed fall is latched (15 s)

Source: [13].

Table A.3: Glove telemetry message (device_id HEALTH_GLOVE_01)

Field group	Content	Used by
Identity	device_id, timestamp	All
Vital signs	heart_rate, SpO2, skin temperature	C1, C3, C4, correlation engine
ECG	Buffer of AD8232 samples at 250 Hz	C1
Motion	MPU6050 acceleration and angular velocity (50 Hz)	C4 (tremor, SOS)
GSR	Smoothed GSR value (50 Hz sampling)	C3
Location	GPS latitude and longitude	Emergency alerts

Source: Project source code and individual theses.

Table A.4: Screening output returned by each engine

Field	Type	Meaning
diagnosis	string	Label, for example VENTRICULAR ECTOPIC BEAT or ECG WARMING UP
severity	integer	0 normal, 1 warning, 2 serious, 3 fall
likelihood_percent	number	Probability of the abnormal condition, 0 to 100
status	string	ready or warming_up; only ready engines enter the risk score

Source: Project source code.

Table A.5: Alert object

Field	Example	Meaning
diagnosis	FALL DETECTED; SUSTAINED BRADYCARDIA	What happened
severity	3	Shared severity scale
trigger	FALL; LOW_HR; HIGH_HR	Rule or event that raised the alert
value	Impact Confirmed	Supporting value shown to carers

Source: Project source code.

Table A.6: Firebase Realtime Database paths

Path	Written by	Content
/processed/HEALTH_GLOVE_01	Hub	Screening results (for example screenings.stress), alerts and vitals
users/{user_id}/telemetry	Hub	vitals, alerts, disease flags, fall_detected, timestamp

Path	Written by	Content
Hub configuration	Hub	Current hub address for device discovery
Facial stress request and result	App and hub	Base64 image, pending flag, device identifier; class, confidence, bounding box

Source: Project documentation and individual theses.

Appendix B Team contribution declaration

The members of Group R26-IT-108 agree that the individual and shared contributions to the project are as recorded in Table B.1.

Table B.1: Agreed individual and shared contributions

Member	Individual contribution	Shared contribution
Abeysinghe S.A.M.M. (IT22081148)	C1 arrhythmia screening: ECG feature extraction, Random Forest classifier, heart-rate rules, posture-aware suppression, evaluation	Hub backend, sensor fusion, correlation engine, Master Risk Score, dashboard, mobile application, integration testing
Lakshan N K T (IT22196156)	C2 Smart Belt: hardware, firmware, fall detection, posture classification, telemetry, backend fall handling, trials	As above
E.A.P. Peshala (IT22146410)	C3 stress monitoring: GSR acquisition, stress engine, facial stress pipeline, Firebase facial exchange, evaluation	As above
Dissanayaka D.M.S.K. (IT22291196)	C4 respiratory and thermal: hypoxemia, sepsis and Parkinson's tremor models, silent hypoxia and early sepsis rules, SOS gesture	As above

Source: Group.

Table B.2: Confirmation by all members

Name	Student ID	Signature	Date
Abeysinghe S.A.M.M.	IT22081148	|	
Lakshan N K T	IT22196156	|	
E.A.P. Peshala	IT22146410	|	
Dissanayaka D.M.S.K.	IT22291196	|	

Source: Group.

Appendix C Academic integrity and AI use disclosure

Generative AI was used in preparing this report as described in Table C.1, in line with the module policy.

Table C.1: Use of generative AI tools

Tool	Purpose	Verification by the team	Sections affected
Claude (Anthropic)	Structuring the report to the common report template; drafting text from the four individual theses and project documents; drawing Figures 2.1, 3.1, 3.2 and 6.1	The team supplied the theses, code and results, checked every figure and number against the individual theses, and reviewed and edited the text	All chapters, Appendices A and B

Source: Group.

The team remains responsible for the content of this report, including every number, claim and reference.